

$$1.1) \frac{\partial \text{PreSLN}(h)_j}{\partial h_i} = \frac{\partial \sigma(z_j)}{\partial h_i}$$

$$= \frac{\partial \sigma(z_j)}{\partial z_j} \frac{\partial (z_j)}{\partial h_i}$$

$$= \frac{\partial \sigma(z_j)}{\partial z_j} \frac{\|h\| \cdot 0 - h_j (h_i / \|h\|)}{\|h\|^2} \rightarrow \frac{\partial \|h\|}{\partial h_i} = \sqrt{h_1^2 + \dots + h_n^2}$$

$$= \frac{-\partial \sigma(z_j)}{\partial z_j} \frac{h_i h_j}{\|h\|^3}$$

$$= \frac{1}{2} (h_1^2 + \dots + h_n^2)^{-\frac{1}{2}} 2 h_i$$

$$= \frac{h_i}{\|h\|}$$

$$2) \frac{\partial \text{PostSLN}(h)_j}{\partial h_i} = \frac{\partial \frac{\sigma(h_j)}{\|\sigma(h)\|}}{\partial h_i}$$

$$= \frac{\|\sigma(h)\| (0 - \sigma(h_j)) \frac{\partial \|\sigma(h)\|}{\partial h_i}}{\|\sigma(h)\|^2}$$

$$= \frac{-\partial(\sigma(h_j))}{\partial h_i} \frac{\sigma(h_j) \sigma(h_i)}{\|\sigma(h)\|^3}$$

$$\frac{\partial \|\sigma(h)\|}{\partial h_i}$$

$$= \frac{1}{2} (\sigma(h_1)^2 + \dots + \sigma(h_n)^2)^{-\frac{1}{2}} 2 \sigma(h_i) \frac{\partial(\sigma(h_i))}{\partial h_i}$$

$$= \frac{\sigma(h_i)}{\|\sigma(h)\|} \frac{\partial(\sigma(h_i))}{\partial h_i}$$

$$3) h = [-1^{=h_1}, 2^{=h_2}]^T$$

$$\Rightarrow r(h) = [0, 2]^T$$

$$z = \left[\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right]^T$$

$$\text{where } i=1 \quad j=2$$

$$\text{Then } \frac{\partial \text{Pre SLN}(h)_2}{\partial h_1} = -1 \frac{-1(2)}{(\sqrt{5})^3} = \frac{2}{(\sqrt{5})^3}$$

$$\text{However, } \frac{\partial \text{Post SLN}(h)_2}{\partial h_1} = \frac{-(0)(0)(0)}{(\sqrt{4})^3} = 0$$

$$1.2 \quad 1) \frac{\partial}{\partial \phi} \int p_x(x) \log(D_\phi(x)) dx + \int p_o(x) [\log(1 - D_\phi(x))] dx$$

$$\Rightarrow p_x(x) \frac{1}{D_\phi(x)} = p_o(x) \frac{1}{1 - D_\phi(x)}$$

$$p_x(x) (1 - D_\phi(x)) = p_o(x) D_\phi(x)$$

$$p_x(x) - p_x(x) D_\phi(x) = p_o(x) D_\phi(x)$$

$$p_x(x) - p_x(x) D_\phi(x) - p_o(x) D_\phi(x) = 0$$

$$D_\phi(x) = \frac{p_x(x)}{p_x(x) + p_o(x)}$$

2)

$$D\phi(x) = \frac{p_x(x)}{p_x(x) + p_x(x)}$$

$$= \frac{1}{2}$$

so the discriminator

just outputs $1/2$ no matter what

3)

$$\frac{2}{2G} \int P_X(x) \log \left(\frac{P_X(x)}{P_X(x) + P_G(x)} \right) + \int P_G(x) \log \left(\frac{P_G(x)}{P_X(x) + P_G(x)} \right)$$

$$\int P_X(x) \log \left[\frac{P_X(x)}{P_X(x) + P_G(x)} \right] + \int P_G(x) \log \left[\frac{P_G(x)}{P_X(x) + P_G(x)} \right] - 2 \ln 2$$

These are both KL Divergences

so are minimized at 0

and when $P_G(x) = P_X(x)$ we have

$$\int P_X(x) \log \left[2 \left(\frac{1}{2} \right) \right] dx + \int P_G(x) \log \left[2 \left(\frac{1}{2} \right) \right] dx - 2 \ln 2$$

$$= -2 \ln 2$$

so the minimum is achieved when $P_G(x) = P_X(x)$
and equals $-2 \ln 2$

$$1.3 \quad \min_{\Theta} -E_{z \sim q_{\phi}(z|x)} [\log p_{\theta}(x|z)] = \min_{\Theta} -E_z \left[\log \left[\frac{1}{(2\pi)^{K/2} |\mathbf{I}|^{1/2}} \exp \left(-\frac{1}{2} (x - D_{\theta}(z))^T \mathbf{I}^{-1} (x - D_{\theta}(z)) \right) \right] \right]$$

$$= \min_{\Theta} -E_z \left[\log \left[\frac{1}{(2\pi)^{K/2} |\mathbf{I}|^{1/2}} \right] \right] = E_z \left[-\frac{1}{2} (x - D_{\theta}(z))^T \mathbf{I}^{-1} (x - D_{\theta}(z)) \right]$$

$$= \min_{\Theta} E_{z \sim q_{\phi}(z|x)} \left[\|x - D_{\theta}(z)\|_2^2 \right], \text{ first term is constant w.r.t. } \Theta$$

1.4

1) When $\frac{c}{n} = 0.01$ and $d=2$

$$f = 1 - (1 - 0.01)^2 = 1 - 0.9801 \\ = 0.0199$$

$$d=10$$

$$f = 1 - (1 - 0.01)^{10} = 1 - 0.9043 \\ = 0.0957$$

$$d=500$$

$$f = 1 - (1 - 0.01)^{500} = 1 - 0.00657 \\ = 0.9934$$

2) for $r = 10$ and $d = 2$

$$\frac{\pi 5^2}{\pi 10^2} = \frac{25}{100}$$

$$d = 10$$

$$\frac{5^{10}}{10^{10}} = \left(\frac{1}{2}\right)^{10} = 0.0009765$$

$$d = 500$$

$$\frac{5^{500}}{10^{500}} = \left(\frac{1}{2}\right)^{500} = 3.05 \times 10^{-151}$$

3) Most points would be near the surface of the sphere

But for VAE a multivariate gaussian has points concentrated in the origin which may not be realistic to points from data and decoder might then generate poor results

But GANs don't approximate a distribution so they don't suffer from this they just learn a mapping